Mahdi Islam

Website | Google Scholar | mahdiiut079@gmail.com | GitHub | LinkedIn

EDUCATION

Erasmus Mundus Joint Master: Medical Imaging & Applications (MAIA) September 2023 – September 2025
University of Girona (Spain) | University of Cassino (Italy) | University of Bourgogne (France) **Grade: 8.96/10.00**
Relevant Courses: Advanced Image Analysis, Machine & Deep Learning, Medical Image Segmentation & Applications, Medical Image Registration & Applications, Computer-Aided Diagnosis
Thesis: Uncertainty-Guided Active Learning for Access Route Segmentation and Planning in Transcatheter Aortic Valve Implantation

Bachelor of Science: Electrical & Electronic Engineering January 2018 – May 2022
Islamic University of Technology (IUT), Bangladesh **Grade: 3.63/4.00**

RESEARCH & WORK EXPERIENCE

Lecturer, ***Department of Electrical & Electronic Engineering***
Independent University, Bangladesh (IUB) January 2026 – Present

- Teaching undergraduate courses in Electrical & Electronic Engineering; supervising student research projects and mentoring thesis students toward research publications.

Research Supervisor, ***Biomedical Instrumentation and Signal Processing Lab Wing***
Center for Computational & Data Sciences (CCDS), Independent University, Bangladesh (IUB) January 2026 – Present

Research Assistant, ***Department of Radiology***
Medical University of Innsbruck, Austria February 2025 – July 2025

- Implemented active learning strategies for an annotation-efficient 3D segmentation pipeline with automated vessel diameter measurement, extracting iliac diameter profiles for access-route planning in Transcatheter Aortic Valve Implantation (TAVI).
- Developed a multimodal pipeline combining 3D deep learning, segmentation-based imageomics, and clinical data from pre-TAVI cardiac CT to predict post-TAVI mortality and five further outcomes across a three-centre cohort, with interpretability via SHAP and Kaplan–Meier analysis.

Lecturer, ***Department of Computer Science & Engineering***
Metropolitan University, Bangladesh August 2022 – June 2023

- Taught core CS/EEE courses; introduced machine learning concepts and mentored sophomore students, increasing student engagement with thesis-based research.

Undergraduate Research Assistant, ***Department of Electrical & Electronic Engineering***
Islamic University of Technology (IUT), Bangladesh April 2021 – April 2022

- Built a novel dataset of plantar-pressure heat-map videos acquired from foot-sole sensors, expanded through image augmentation.
- Developed a real-time multiclass CNN classifier for gait anomaly detection using Keras and TensorFlow (published at ICBIR 2022).

PUBLICATIONS

Published / Presented

[1] **M. Islam**, M. Tabassum. *"Brain Metastases Segmentation for BraTS 2026 Task 1: A Multi-Architecture Comparison."* **BraTS 2026 (BraTS-METS), MICCAI 2026**, October 2026 [Code] | [Preprint] | Challenge Paper | Poster | **9th place**

[2] E. Almar-Muñoz, **M. Islam**, M. Tabassum, C. Kremser, M. Haltmeier, A. Mayr. *"Uncertainty-Guided Active Learning for Access Route Segmentation and Planning in Transcatheter Aortic Valve Implantation."* **ISMRM-ISMRT Annual Meeting & Exhibition**, Cape Town, South Africa, May 2026 [Abstract #560-03-010] | Conference Abstract | Digital Poster

[3] J. Bai, Y. Tang, Z. Zhou, **M. Islam**, M. Tabassum, E. Almar-Muñoz, *et al.* *"FUGC: Benchmarking Semi-Supervised Learning Methods for Cervical Segmentation."* **IEEE Transactions on Medical Imaging**, 2026 [DOI] | Journal Article

[4] **M. Islam**, M. Tabassum, A. Mayr, C. Kremser, M. Haltmeier, E. Almar-Muñoz. *"Uncertainty-Guided Active Learning for Access Route Segmentation and Planning in Transcatheter Aortic Valve Implantation."* **Journal of Imaging (MDPI)**, vol. 11, no. 9, art. 318, 2025 [DOI] | Journal Article

[5] **M. Islam**, M. Tabassum, M. Elbatel, A. Mayr, C. Kremser, M. Haltmeier, E. Almar-Muñoz. *"Semi-Supervised Transformer-Based Cervical Segmentation: FUGC 2025 Challenge."* **IEEE 22nd International Symposium on Biomedical Imaging (ISBI), Challenge Track**, 2025 [DOI] | Challenge Report | Challenge Presentation | **5th place**

[6] **M. Islam**, M. Tabassum, M. M. Nishat, F. Faisal, M. S. Hasan. *"Real-Time Clinical Gait Analysis and Foot Anomalies Detection Using Pressure Sensors and Convolutional Neural Network."* **International Conference on Business and Industrial Research (ICBIR)**, pp. 717–722, 2022 [DOI] | Conference Paper

Under Review / In Preparation

[7] E. Almar-Muñoz, **M. Islam**, M. Tabassum, M. Haltmeier, M. Pamminger, L. Neumann, C. Kremser, M. Reindl, S. J. Reinstadler, B. Metzler, A. Mayr. *"Automated Aortoiliofemoral Segmentation and Transfemoral Access Planning Using Non-Contrast MR Angiography for Transcatheter Aortic Valve Replacement Guidance."* **Radiology: Cardiothoracic Imaging** [Under Review] | Journal Article

[8] E. Almar-Muñoz*, M. Tabassum*, **M. Islam***, C. Dlaska, C. Kremser, M. Haltmeier, M. Pekar, M. Kantor, F. Sturla, S. Saitta, R. Gorla, A. Varga-Szemes, J. Osorio-Velasquez, A. Mayr. (*equal contribution) *"An Imageomics and Deep Learning Approach for Predicting Transcatheter Aortic Valve Implantation Outcomes."* **Manuscript** [In Preparation] | Journal Article

PROJECTS

LLM-Guided Feature Selection for Drug Response Prediction | *Link*
Independent Research Project August 2026 – Present

- Built an end-to-end pipeline predicting cancer cell-line drug sensitivity (LN_IC50) from ∼28,000-gene expression profiles under a small-model constraint.
- Compared variance-based feature selection against LLM-guided selection informed by drug mechanism and pathway knowledge, keeping the LLM blind to expression and response data to prevent leakage.
- Evaluated across 25 drugs and multiple seeds with paired Wilcoxon signed-rank tests; TabPFN consistently and significantly outperformed XGBoost, with performance still improving at the largest feature budget tested.

Curvature-Aware Parameter-Efficient Fine-Tuning for Vision–Language Models | *Link*
Remote Research Collaboration July 2025 – Present

- Extended LoRA with Fisher-information-based (K-FAC-style) curvature approximations to stabilize low-rank adaptation in large VLMs.
- Implemented adaptive rank reprojection and rank-stabilized LoRA to dynamically control model capacity under limited supervision.

Multimodal Outcome Prediction for Post-TAVI Mortality Using Cardiac CT and CMR | *Link*
Medical University of Innsbruck (remote collaboration) March 2025 – July 2026

- Built an outcome-prediction pipeline using cardiac CT and CMR data to estimate post-TAVI mortality risk.
- Extracted segmentation-based imageomics features; applied SHAP and correlation analysis to identify anatomical predictors of adverse outcomes.
- Explored self-supervised pretraining (VoCo, masked autoencoders) to learn domain-specific representations from limited labeled data.

Deep Learning-Based Brain Tissue Segmentation Using U-Net and MRI Data | *Link*
University of Girona November 2024 – January 2025

- Developed a U-Net model for CSF, GM, and WM segmentation from IBSR18 MRI scans with tailored preprocessing and augmentation.
- Built a PyTorch training pipeline with Weights & Biases for experiment tracking and hyperparameter optimization.

4D Chest CT Volume Registration: DIR-Lab Challenge | *Link*
University of Girona November 2024 – January 2025

- Developed a baseline deformable registration model using elastix for thoracic CT alignment across respiratory phases in COPD patients.
- Built a VoxelMorph-based deep learning pipeline to improve registration accuracy and benchmarked it against the baseline.

Skin Cancer Detection from Dermoscopic Images: Comparing ML & DL Approaches | *Link*
University of Girona September 2024 – January 2025

- Built end-to-end ML and DL pipelines for binary and multiclass skin lesion classification, including preprocessing (hair removal, color normalization, ROI extraction) and handcrafted features (GLCM, LBP, Gabor, HOG).
- Implemented SVM, Random Forest, and XGBoost classifiers alongside EfficientNet-based models, addressing class imbalance through resampling.

Atlas and Tissue-Model Guided Gaussian Mixture Models for Brain Tissue Segmentation | *Link*
University of Girona September 2024 – November 2024

- Developed a hybrid segmentation approach combining Gaussian Mixture Models with probabilistic atlases, integrating spatial and intensity information for GM, WM, and CSF segmentation from MRI.
- Implemented multiple Expectation-Maximization initialization strategies (K-means, tissue model), improving Dice scores over traditional methods.

Colorectal Cancer Tissue Classification and Gland Segmentation from Histopathology Images | *Link*
University of Cassino March 2024 – May 2024

- Built a classical segmentation pipeline (K-means, Watershed) and an ML classification pipeline using GLCM, LBP, and Gabor features with XGBoost, LightGBM, and SVM.
- Engineered a PyTorch deep learning segmentation pipeline with UNet/UNet++ and VGG16, ResNet, and EfficientNet backbones.

ACCOMPLISHMENTS

BraTS 2026 Cluster of Challenges | MICCAI 2026 October 2026

- Ranked **9th internationally** among 689 competing teams.

Fetal Ultrasound Grand Challenge (FUGC) | IEEE ISBI 2025 | *Certificate* March 2025

- Ranked **5th internationally** in this IEEE ISBI medical image segmentation challenge.

Intherapy Graduate School Scholarship | University of Bourgogne, France | *Certificate* January 2025
Associate Data Scientist Certificate | DataCamp | *Certificate* November 2024

EXTRACURRICULAR ACTIVITIES

Coordinator – IT & Web Design, Erasmus Mundus Association Bangladesh January 2026 – Present
Deputy Coordinator – IT & Web Design, Erasmus Mundus Association Bangladesh January 2025 – December 2025
Captain, IUT University Tennis Club April 2021 – May 2022

SKILLS

Programming: Python, R, MATLAB, C++, SQL, JavaScript | **DL Architectures:** CNNs, Transformers, LLMs, VLMs, LSTMs, RNNs
Frameworks: PyTorch, MONAI, TensorFlow, Hugging Face, Scikit-learn | **Medical Tools:** ITK-SNAP, 3D Slicer, FreeSurfer, ITK-Elastix, SPM
Data & Visualization: Pandas, NumPy, Matplotlib, Seaborn, OpenCV, Scikit-image | **Compute:** Distributed Training (DDP), Multi-GPU HPC Clusters

RESEARCH INTERESTS

Multimodal Data Fusion, Medical Image Segmentation & Analysis, Digital & Computational Pathology, Self-Supervised & Semi-Supervised Learning, Parameter-Efficient Fine-Tuning of Foundation Models, Radiomics & Biomarker Analysis for Early Disease Detection, Prognosis, and Individualized Risk Prediction, Longitudinal Modeling.

STANDARDIZED TEST SCORES

IELTS: Listening 8.5 | Reading 9 | Speaking 8 | Writing 7 | **Overall 8**

REFERENCES

Dr. Agnes Mayr, Associate Professor, PD | Master's Thesis Supervisor
Department of Radiology
Medical University of Innsbruck, 6020 Innsbruck, Austria
Email: a.mayr@i-med.ac.at
Dr. Alessandro Bria, Associate Professor | Course Instructor and Project Supervisor
Department of Electrical and Information Engineering
University of Cassino and Southern Latium, 03043 Cassino, Italy
Email: a.bria@unicas.it